

NAMIBIA SENIOR SECONDARY CERTIFICATE

BIOLOGY ORDINARY LEVEL

6116/1

PAPER 1 Multiple Choice

45 minutes

Marks 40

2024

Additional Materials: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Write in soft pencil.
- Answer **all** questions.
- Make sure that you receive the multiple choice answer sheet with your **Candidate Number** on it.
- There are **forty** questions on this paper.
- For each question, there are four possible answers **A, B, C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.
- If you want to change an answer, thoroughly erase the one you wish to delete.

Read the instructions on the answer sheet carefully.

- Each correct answer will score one mark.
- All questions in this paper carry equal marks.
- Any rough working should be done in this booklet.
- You may use a non-programmable calculator.

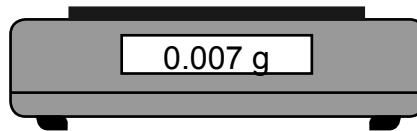
This document consists of **15** printed pages and **1** blank page.



Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 The diagram shows an instrument that can be used in a Biology laboratory.



What does the instrument measure?

- A mass
 - B temperature
 - C time
 - D volume
- 2 Which two taxa are used in naming organisms using the binomial system?
- A family and kingdom
 - B family and genus
 - C genus and species
 - D kingdom and species
- 3 Palisade cells are found in plant leaves.

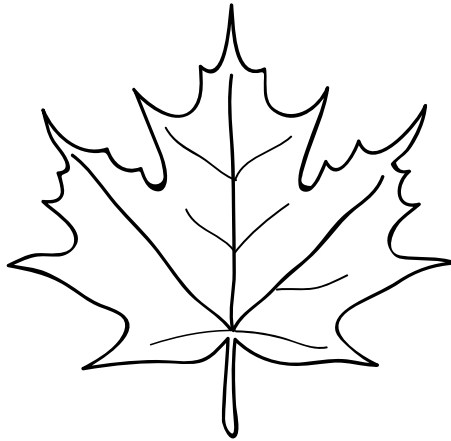
Which feature is present in a palisade cell but **not** in a liver cell?

- A cell membrane
 - B chloroplast
 - C cytoplasm
 - D nucleus
- 4 The actual size of an object is 2000 μm . A drawing of the same object measures 50 mm.

What is the magnification of the drawing?

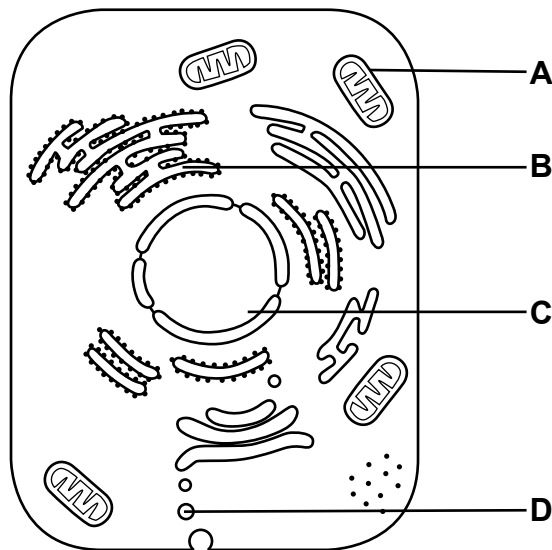
- A $\times 0.025$
- B $\times 25$
- C $\times 100$
- D $\times 100\,000$

- 5 The diagram shows a leaf.



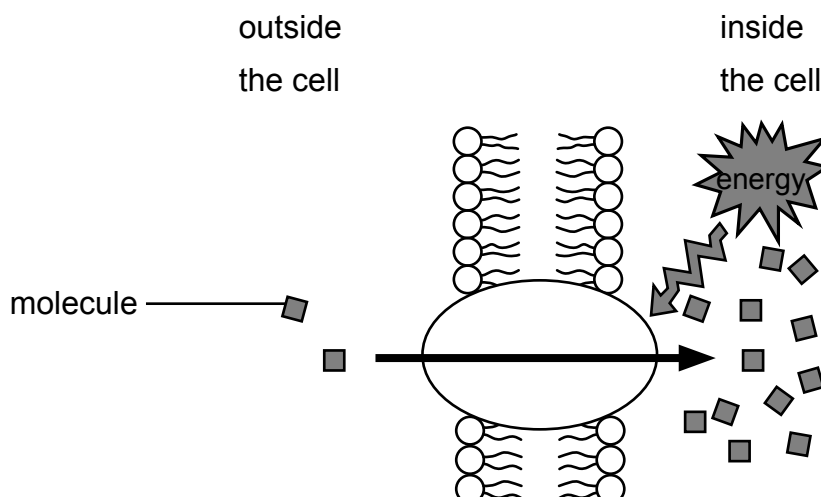
At what level of organisation is a leaf?

- A cell
 - B organ
 - C organ system
 - D tissue
- 6 The diagram shows a eukaryotic cell.



Which structure produces ATP during respiration?

- 7 The diagram shows the movement of molecules from outside the cell into the cell.



Which phrase describes how the molecules move from outside the cell into the cell?

- A** against the concentration gradient by active transport
 - B** against the concentration gradient by facilitated diffusion
 - C** down a concentration gradient by osmosis
 - D** down a concentration gradient by diffusion
- 8 DNA bases pair up to form two strands during DNA replication.

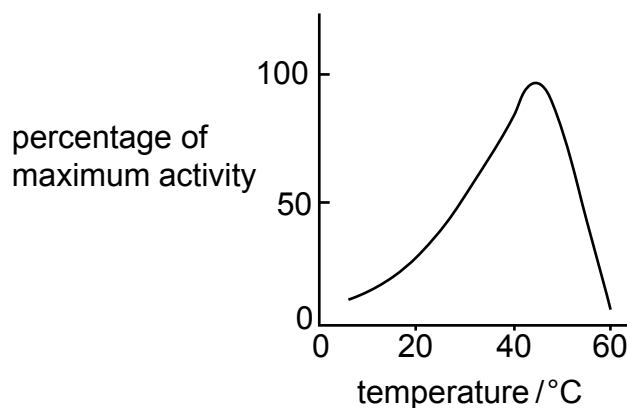
Which bases are paired correctly?

- A** G with A
 - B** G with C
 - C** G with G
 - D** G with T
- 9 A learner carried out a food test to determine whether milk sugar, lactose, is a reducing sugar.

Which row is a correct description of the test?

	reagent to be used	heating required	positive result
A	Benedict's solution	no	orange colour
B	biuret solution	yes	blue colour
C	Benedict's solution	yes	brick red colour
D	biuret solution	no	blue colour

- 10 The graph shows the effect of temperature on the activity of an enzyme.



At which temperature is the enzyme **most** active?

- A 15°C
 - B 35°C
 - C 45°C
 - D 55°C
- 11 What will happen to a green plant grown in soil that is deficient in nitrate ions?
- A It will have large leaves and small flowers.
 - B It will have purple leaves and poor root growth.
 - C It will have yellow leaves and poor overall growth.
 - D It will have white leaves and a thick stem.
- 12 In which part of the human alimentary canal does the digestion of starch begin?
- A mouth
 - B oesophagus
 - C small intestine
 - D stomach
- 13 In which order do these events occur in human nutrition?
- A digestion → ingestion → absorption → egestion
 - B digestion → ingestion → egestion → absorption
 - C ingestion → digestion → absorption → egestion
 - D ingestion → digestion → egestion → absorption

14 The information below is taken from four packets of food.

Which food would be suitable for a person suffering from constipation?

A

energy	163 kJ
protein	0.1 g
fat	0 g
carbohydrates	9.4 g
fibre	1 g

B

energy	466 kJ
protein	1.5 g
fat	0.4 g
carbohydrates	27 g
fibre	4.9 g

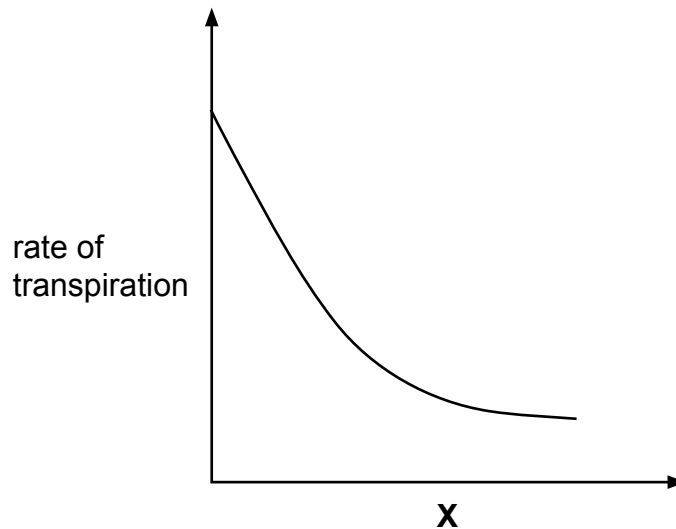
C

energy	1054 kJ
protein	19 g
fat	7.3 g
carbohydrates	27 g
fibre	8.1 g

D

energy	458 kJ
protein	2.1 g
fat	3.3 g
carbohydrates	19 g
fibre	1.1 g

15 The graph shows how the rate of transpiration is affected by factor **X**.

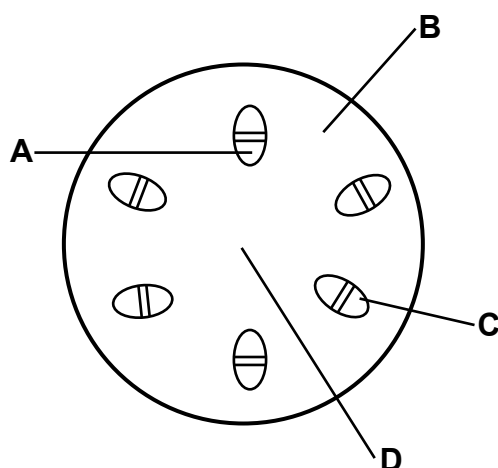


What could be factor **X**?

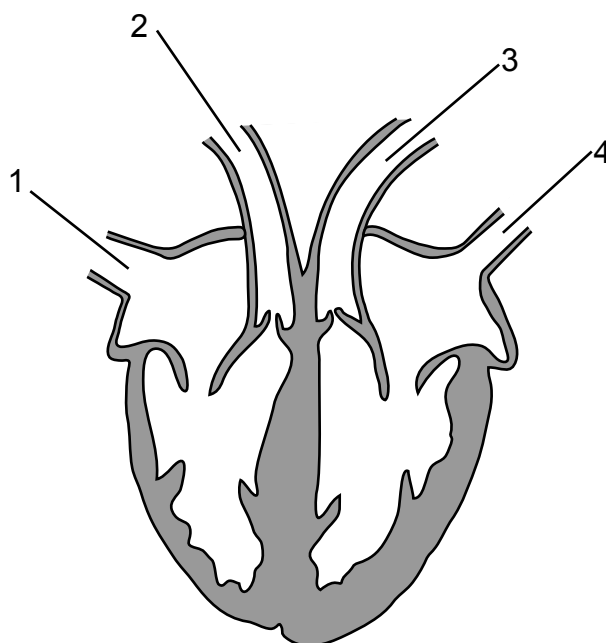
- A** humidity
- B** light intensity
- C** temperature
- D** wind speed

16 The diagram shows a section through a plant stem.

In which tissue does the translocation of systemic pesticides take place?



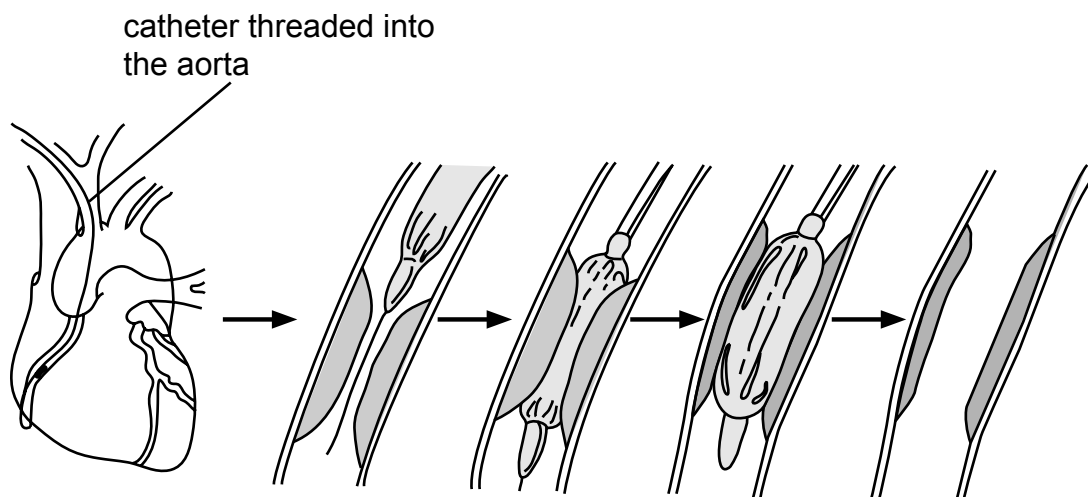
17 The diagram shows a vertical section through the heart.



What are the functions of the blood vessels numbered 1, 2, 3 and 4?

	carries blood to body	carries blood to lungs	carries blood from lungs	carries blood from body
A	1	4	3	2
B	2	1	3	4
C	2	3	4	1
D	3	2	4	1

18 The diagram shows a method of treating coronary heart disease.



Which method of treating a coronary heart disease is shown in the diagram?

- A** angioplasty
- B** coronary by-pass
- C** stent
- D** using aspirin

19 The body has different types of defences against pathogens.

- 1 antibodies
- 2 hair in the nose
- 3 mucus
- 4 skin

Which defences are mechanical?

- A** 1 and 2 only
- B** 1, 2 and 3 only
- C** 2, 3 and 4 only
- D** 2 and 4 only

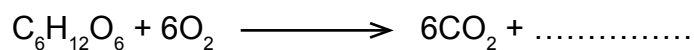
- 20** The table shows the composition of inspired and expired air.

	inspired air/%	expired air/%
carbon dioxide	0.04	X
oxygen	21	16
nitrogen	78	Y
other gases	1.0	1.0

What are the percentages of **X** and **Y**?

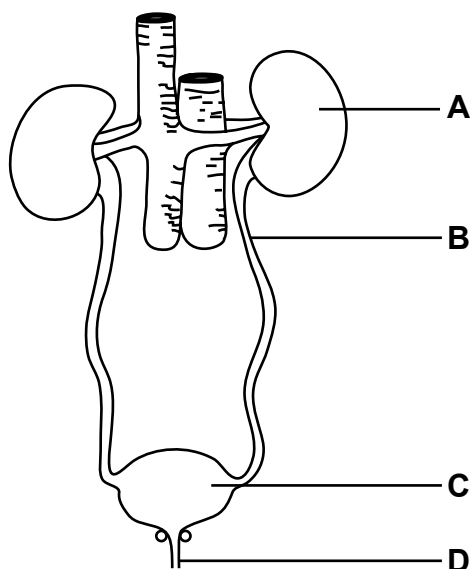
	X	Y
A	0.04	21
B	4	78
C	0.04	78
D	4	21

- 21** The equation for aerobic respiration given below is incomplete.



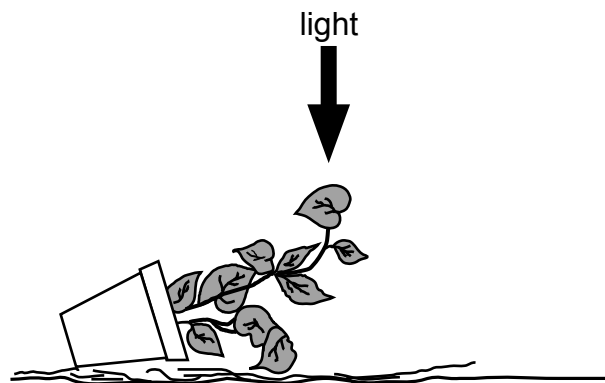
Which information is needed to complete it?

- A** $\text{C}_3\text{H}_6\text{O}_3$
B $2\text{C}_2\text{H}_5\text{OH}$
C $6\text{H}_2\text{O}$
D 6O_2
- 22** The diagram shows the human urinary system.



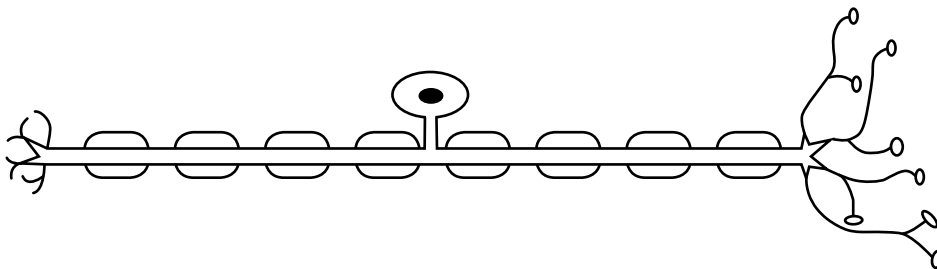
Which organ re-absorbs glucose so that glucose is not lost in urine?

- 23** In which way is a kidney transplant advantageous compared to the dialysis treatment?
- A** there is no organ rejection
 - B** it is a permanent solution
 - C** it is less expensive
 - D** it is not a risky procedure
- 24** The diagram shows a seedling responding to light.



What explains the response of the shoot?

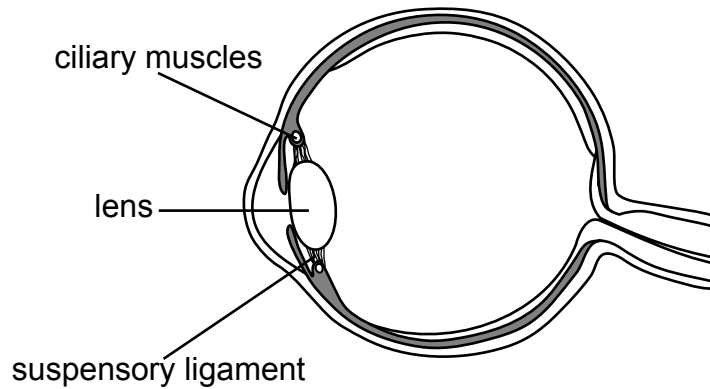
- A** inhibited cell elongation of the lower side of the shoot
 - B** equal growth of cells on the upper and lower sides of the shoot
 - C** stimulated cell elongation of the lower side of the shoot
 - D** slowed growth of cells of the whole shoot
- 25** The diagram shows a neurone.



Which neurone is shown in the diagram?

- A** motor
- B** relay
- C** sensory
- D** synapse

26 The parts of the eye shown in the diagram are involved in accommodation.



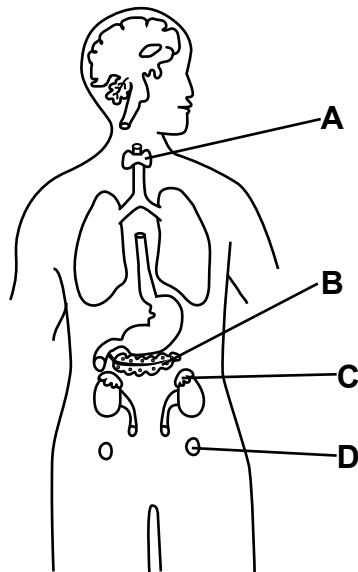
How does each part change to accommodate a near object?

	ciliary muscle	suspensory ligament	lens
A	contracts	slackens	more convex
B	contracts	tightens	less convex
C	relaxes	tightens	more convex
D	relaxes	slackens	less convex

27 Which response is an example of a voluntary action?

- A** changing the size of the pupil when entering a brightly lit room
- B** salivating when looking at a delicious meal
- C** answering the teacher's question in class
- D** withdrawing a hand from a hot object

28 The diagram shows the major endocrine glands in the human body.



Which endocrine gland produces insulin?

29 Below is a list of processes involved in temperature regulation in humans.

shivering sweating vasoconstriction vasodilation

Which processes occur when the body temperature rises above 37°C?

- A** shivering and sweating
- B** sweating and vasoconstriction
- C** sweating and vasodilation
- D** vasoconstriction and vasodilation

30 The diploid number of chromosomes in the nucleus of the cell of a domestic cat is 38.

How many chromosomes will be in a sperm cell nucleus and a skin cell nucleus?

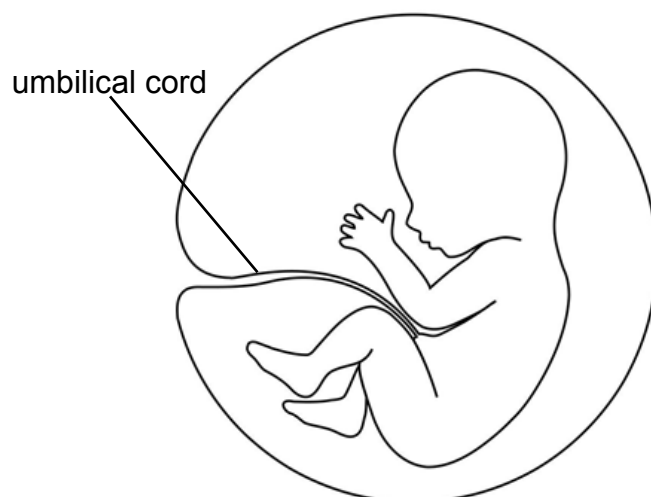
	sperm cell nucleus	skin cell nucleus
A	19	19
B	19	38
C	38	38
D	38	19

31 The menstrual cycle is controlled by the hormones FSH, LH, oestrogen and progesterone.

Which row shows a correct match between a hormone and its function?

	hormone	function
A	FSH	repair the endometrium after menstruation
B	LH	causes the release of the mature egg from the ovary
C	oestrogen	maintains the thickness of the endometrium
D	progesterone	causes the growth of the follicles in the ovary

32 The diagram shows a foetus during pregnancy.



What is the function of the umbilical cord?

- A** absorbs mechanical shock
- B** absorbs excess heat during pregnancy
- C** attaches the foetus to the placenta
- D** produces hormones during pregnancy

33 Which method of birth control involves surgery?

- A** abstinence
- B** condom
- C** IUD
- D** vasectomy

34 Height in plants is controlled by a pair of alleles. Allele **T** for tall height is dominant over allele **t** for short height. Plants can be cross-pollinated to produce the required height.

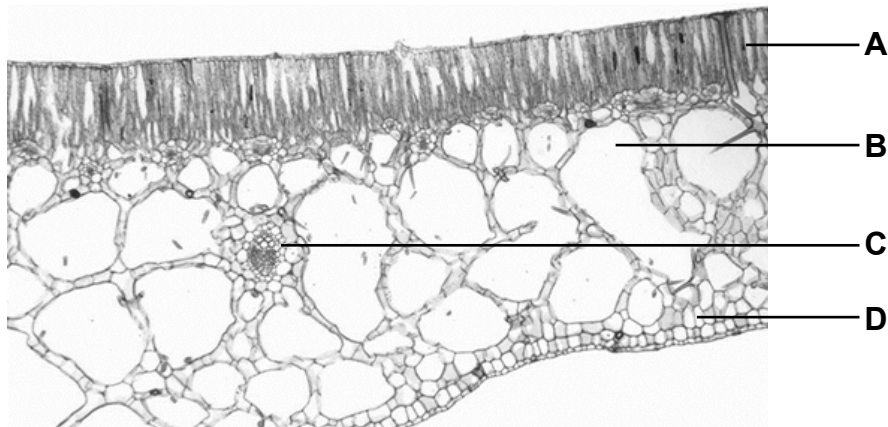
Which crossing will produce a phenotypic ratio of 3:1 tall to short in the offspring?

- A** $TT \times tt$
- B** $Tt \times tt$
- C** $Tt \times Tt$
- D** $TT \times Tt$

35 Which type of variation is discontinuous?

- A** height in humans
- B** length of leaves
- C** mass of seeds
- D** sex in humans

36 The diagram shows a photomicrograph of a leaf of a hydrophyte.

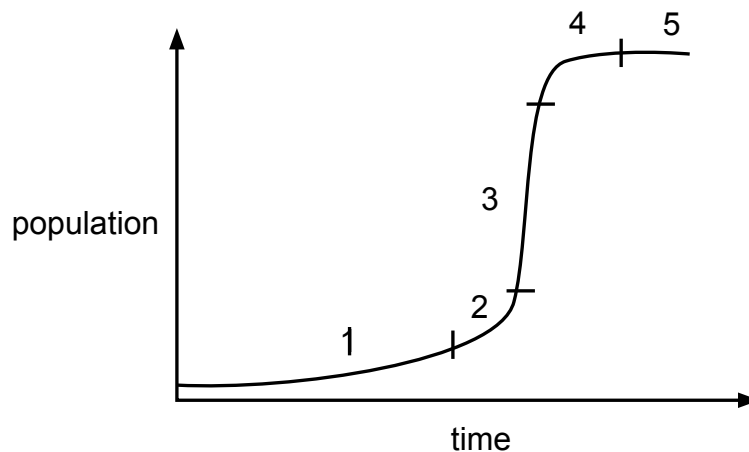


Which adaptive feature enables the leaf to float on the water surface?

37 Which organism has the **lowest** amount of energy available compared to the energy entering the food chain?

- A carnivore that feeds on herbivores
- B herbivores that feed on the producers
- C producer that makes its own food
- D consumer that feeds on carnivores

38 The graph shows the phases of the sigmoid growth curve of population growth.



Which factor could limit the growth of the population at 5?

- A decreased predation
- B lack of diseases
- C limited food supply
- D excess of reproducing individuals

39 The events listed below occur during eutrophication in a lake.

- 1 aerobic bacteria break down dead organic matter
- 2 algae absorb the nutrients, bloom and die
- 3 amphibians, crustaceans and fish die due to lack of oxygen
- 4 nutrients enter the lake
- 5 water become deoxygenated

Which sequence of events is correct?

- A** 2 → 1 → 3 → 4 → 5
B 2 → 1 → 5 → 4 → 3
C 4 → 3 → 5 → 1 → 2
D 4 → 2 → 1 → 5 → 3

40 Which methods can be used to help sustain fish stocks?

key - ✓ method used

✗ method not used

	education	legal quotas	restocking	increased fishing
A	✓	✓	✓	✗
B	✓	✓	✗	✗
C	✓	✗	✓	✓
D	✗	✓	✓	✓

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